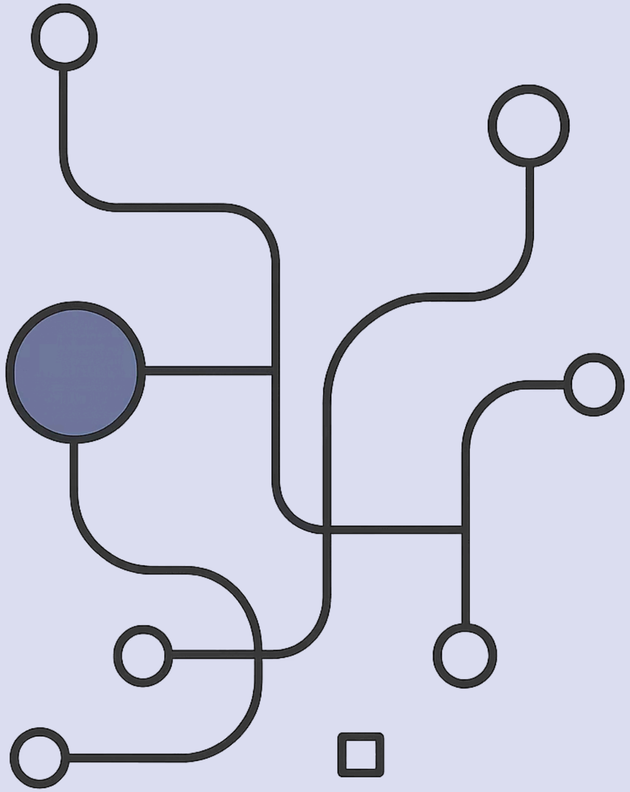
Input Features and Target Variable

A beginner-friendly deep dive into the most important foundation before building any Machine Learning model.



What should the machine learn?

Before building any ML model, we must clearly define:

Input

Features — the information we give the model

Output

Target Variable — what the model must predict

Simple Meaning

Can we predict final marks based on study hours, attendance, and previous marks?

Inputs (Features)

- Study hours
- Attendance
- Previous marks

Output (Target Variable)

- Final marks

Inputs are called **features**. The output is called the **target variable**.

House Price Example

Goal: Predict house price. The model learns: Area + Bedrooms + House Age + Distance → Price

Area	Bedrooms	House Age	Distance	Price
1000 sqft	2	5 yrs	8 km	₹55 lakh
1500 sqft	3	3 yrs	5 km	₹82 lakh
2000 sqft	4	2 yrs	4 km	₹120 lakh
900 sqft	2	10 yrs	12 km	₹42 lakh

Column Roles

Column	Role
Area	Feature
Bedrooms	Feature
House Age	Feature
Distance from City	Feature
Price	Target Variable

Key Distinction

Features are the inputs — the information provided to the model.

The target variable is the answer we want to predict.

What Is a Feature?

A feature is an input column used to make a prediction — useful information given to the model.

Area & Size

Area, bedrooms, bathrooms

Property Details

House age, parking availability

Location

Distance from city, nearby metro

Features Are Clues

Clues (Features)

Area = 1500 sqft · Bedrooms = 3 · Age = 3 yrs · Distance = 5 km

Answer (Target)

Price = ₹82 lakh

The model learns how clues connect to the answer.

What Is a Target Variable?

The target variable is the column we want to predict. For house price prediction, Target = Price.

Also Called

- Label
- Output
- Dependent variable
- y (in code)

The Question It Answers

Whatever your prediction question asks — *"What is the price?"* — that answer column is your target variable.

Target Variable Examples

Problem	Target Variable
Predict house price	Price
Predict student marks	Marks
Predict pass/fail	Result
Predict disease/no disease	Disease Status
Predict salary	Salary
Predict spam/not spam	Spam Status

X and y in Machine Learning

X (Capital)

Input features — multiple columns

Area, Bedrooms, House Age, Distance

y (lowercase)

Target variable — one column

Price

Read it as:

> Given X, predict y.

X is capital because it contains many columns. y is lowercase because it is usually one column.

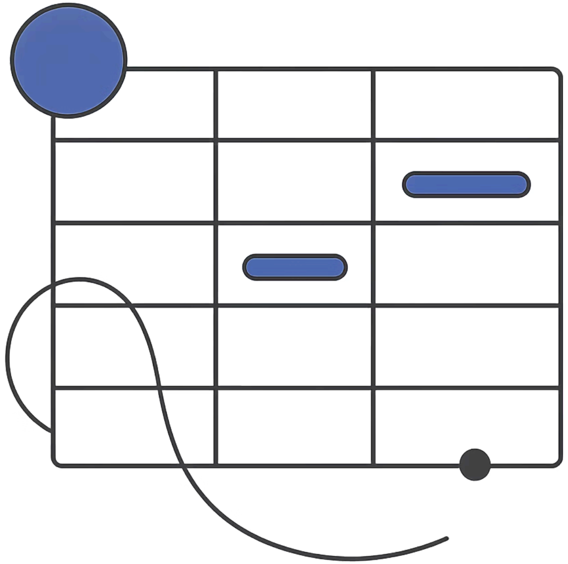
Python Example

The first step before training any model:

```
X = df[["area", "bedrooms", "house_age", "distance"]]
y = df["price"]
```

Use area, bedrooms, house age, and distance to predict price. X holds all input features; y holds the single target column.

Why X Is Capital and y Is Small



X — Multiple Columns

X is a table with many columns: area, bedrooms, age, distance.

y — One Column

y is a single column: price.

The capitalization convention reflects this structural difference.

One Row Example

Area	Bedrooms	House Age	Distance	Price
1500	3	3	5	82

Features (X)	Target (y)
Area = 1500 · Bedrooms = 3 · House Age = 3 · Distance = 5	Price = 82

Many Rows Teach the Model

The model learns from many examples:

01

Row 1

features → target

02

Row 2

features → target

03

Row 3 ... N

features → target

04

New Row

features → predict target

New Prediction Example

A new house arrives — price is unknown:

Area	Bedrooms	House Age	Distance	Price
1700	3	4	6	?

Features are available. Target is unknown. The trained model predicts: Price ≈ ₹90 lakh

Important Rule

GOLDEN RULE

Target should NOT be inside features

❌ Wrong

```
X = df[["area", "bedrooms",
     "house_age", "distance",
     "price"]]
y = df["price"]
```

Why Wrong?

The model is already seeing the answer. It is like giving an exam question with the answer printed beside it.

Correct Feature Selection

```
X = df[["area", "bedrooms", "house_age", "distance"]]
y = df["price"]
```

Golden Rule: Features should never contain the target column.

Doctor Analogy

Features (Symptoms & Tests)

- Age, Weight
- Blood pressure
- Sugar level, Cholesterol
- Symptoms

Target (Diagnosis)

Disease / No Disease

Just as a doctor uses symptoms and test values to decide a diagnosis, the ML model uses features to predict the target.

Student Marks Prediction

Goal: Predict final marks. This is Regression because final marks is a number.

Study Hours	Attendance	Prev. Marks	Final Marks
2	60	40	45
5	85	70	78
6	90	80	88
1	45	25	30

Student Result Classification

Goal: Predict pass or fail. This is Classification because result is a category.

Study Hours	Attendance	Prev. Marks	Result
2	60	40	Fail
5	85	70	Pass
6	90	80	Pass
1	45	25	Fail

How to Identify Features and Target

1	2
What do I want to predict? That is the target variable. Example: I want to predict house price → Target = Price	**What information do I use to predict it?** Those are the features. Example: Features = Area, Bedrooms, Age, Distance

Feature Selection Thinking

Not every column is useful. Suppose house data has:

✅ Useful Features

- Area
- Bedrooms
- Age
- Distance

❌ Not Useful / Risky

- House ID
- Owner Name

Good features must have a relationship with the target.

Features & Targets Across Domains

Domain	Features	Target
Real Estate	Area, bedrooms, location, age	Price
Education	Study hours, attendance, prev. marks	Final marks
Banking	Income, credit score, loan amount	Loan approved/rejected
Healthcare	Age, BP, sugar level, symptoms	Disease status
Retail	Month, product, discount, ad spend	Sales amount
HR	Experience, skills, interview score	Salary

Full Python Example

```
import pandas as pd

data = {
 "area": [1000, 1500, 2000, 900],
 "bedrooms": [2, 3, 4, 2],
 "house_age": [5, 3, 2, 10],
 "distance": [8, 5, 4, 12],
 "price": [55, 82, 120, 42]
}

df = pd.DataFrame(data)

X = df[["area", "bedrooms", "house_age", "distance"]]
y = df["price"]

print(X)
print(y)
```

Common Confusions Cleared

Is target also a column?

Yes — target is a column in training data. But when predicting new data, the target will be unknown.

Can we have multiple features?

Yes — most real ML problems have many features.

Can target be text?

Yes — if the target is text or a category, it is Classification.

Final Summary

Features = X

Input columns — Area, Bedrooms, Age, Distance

Target = y

Output column — Price

Golden Rule

Never include the target column inside features

Quick Quiz

Goal: Predict salary. Identify features and target:

Experience	Skills Score	Interview Score	Salary
1	60	55	₹4 LPA
3	75	70	₹7 LPA
5	85	80	₹12 LPA

Which columns are features? Which column is the target variable?

Quick Quiz Answer

✅ Features (X)

Experience

Skills Score

Interview Score

🎯 Target Variable (y)

Salary

Use experience, skills, and interview score to predict salary.